

AI Tender & Vendor Evaluation Assistant

System Architecture Document - Version 1 Prototype

Prepared For: Internship Review & Mentor Evaluation

Role: Senior Software Architect

Scope: High-Level System Design & AI Pipeline Planning

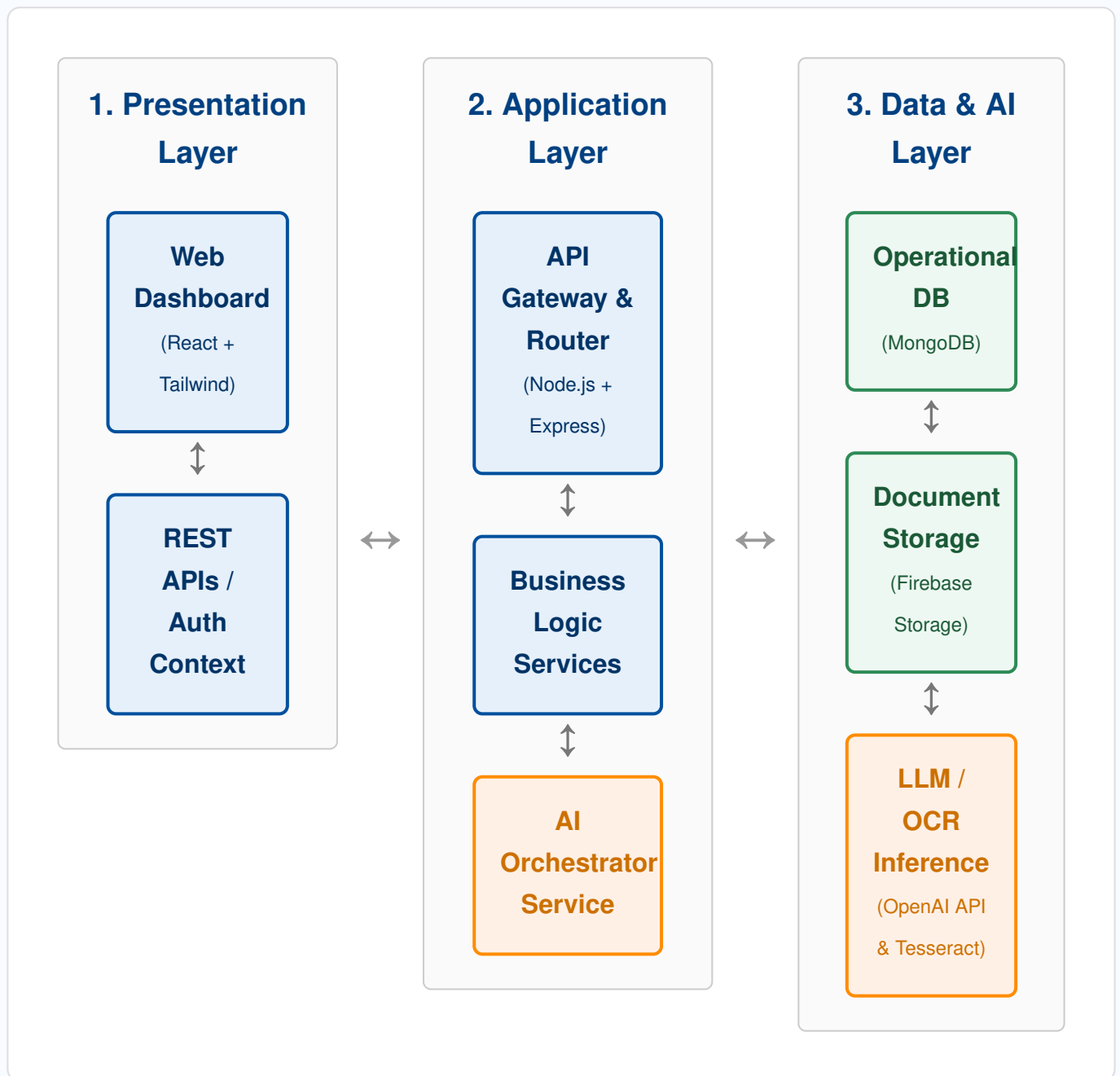
1. Project Overview

The **AI Tender & Vendor Evaluation Assistant** is an enterprise-grade procurement application designed to automate the manual, time-consuming process of tender evaluation. By leveraging artificial intelligence, natural language processing, and advanced document parsing, the system streamlines the procurement lifecycle.

Core Capabilities: Processing uploaded tender specifications and vendor quotations, executing optical character recognition (OCR) on PDF files, extracting vital commercial and technical parameters, verifying compliance against requirements, algorithmically scoring vendors, and generating comprehensive recommendation reports.

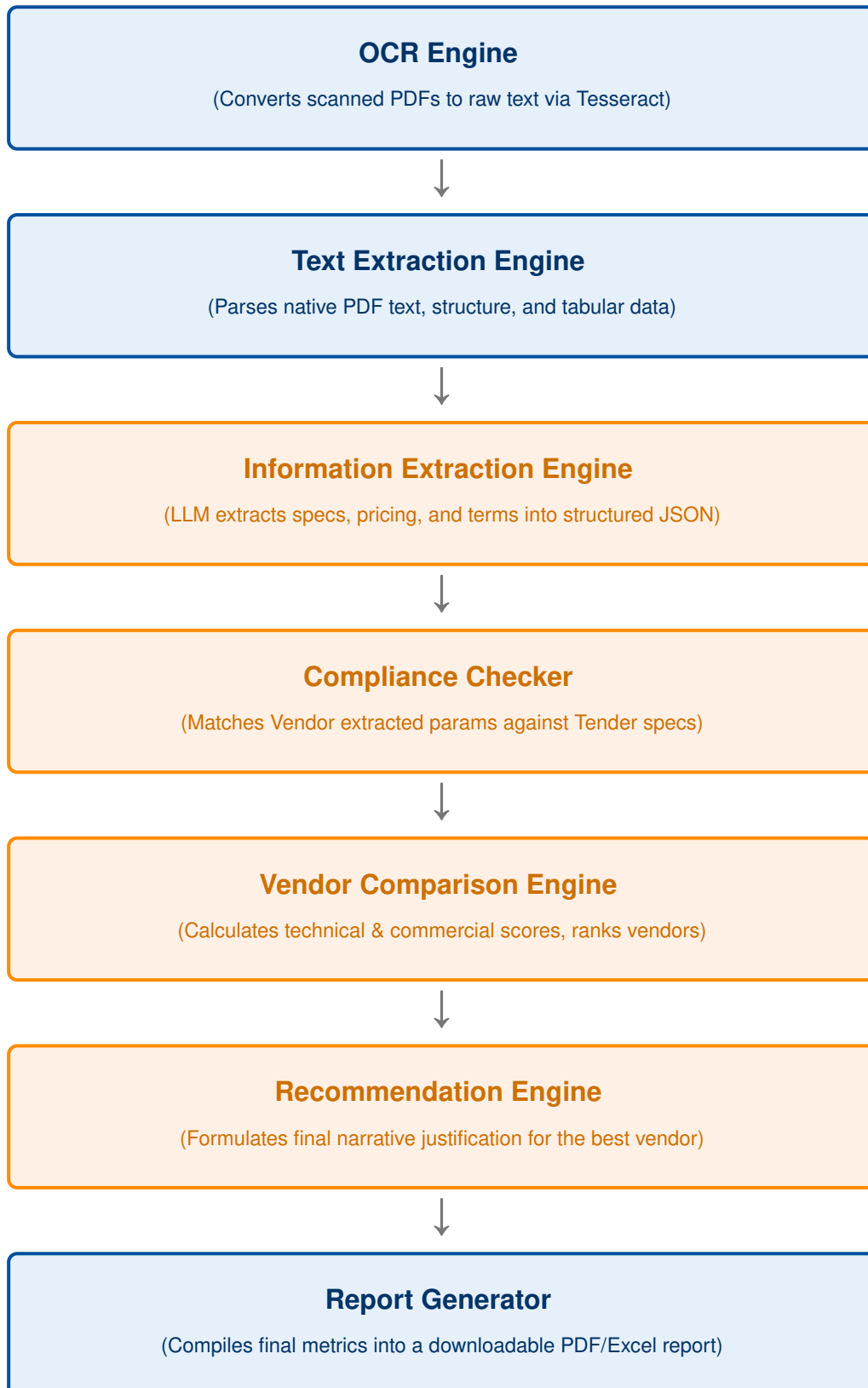
2. High-Level Architecture

The system follows a modern decoupled 3-tier architecture, integrating a dedicated AI processing engine for intensive operations.



3. Component Architecture (AI System Modules)

The AI Engine is broken down into modular micro-components, ensuring separation of concerns and maintainability.



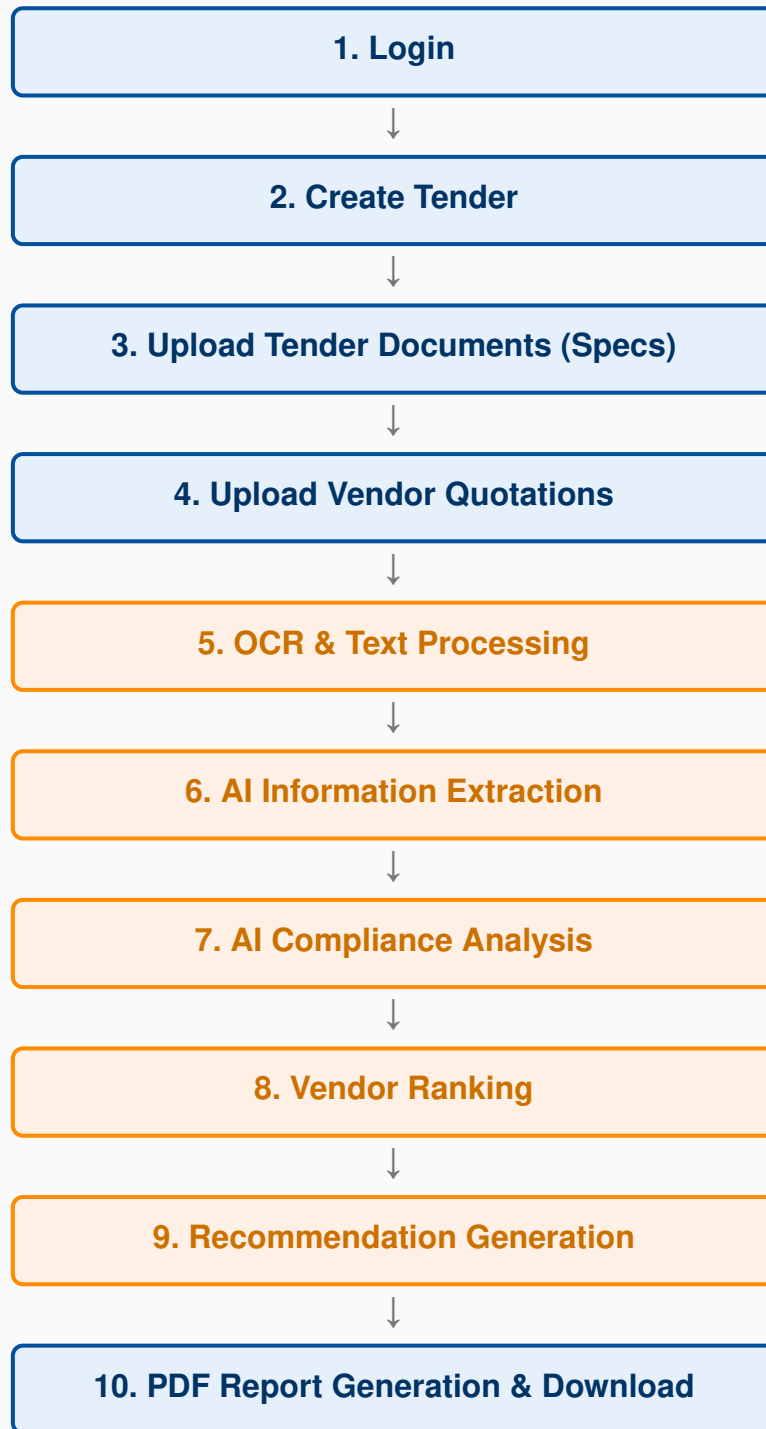
4. Database Design (Entity Relationship)

The operational data is managed in a NoSQL structure with the following core collections and relationships.

Entity / Collection	Core Attributes	Relationships
Users	userId (PK), name, email, role, department	1:N with Tenders
Tenders	tenderId (PK), createdBy (FK), title, deadline, status	1:N with Documents, 1:N with Vendors, 1:1 with Comparisons
Documents	docId (PK), tenderId (FK), vendorId (FK), fileUrl, docType	Belongs to Tender / Vendor
Vendors	vendorId (PK), tenderId (FK), vendorName, submitDate, extractedData (JSON)	Belongs to Tender
Comparisons	comparisonId (PK), tenderId (FK), scoringMatrix (JSON), topVendorId (FK)	1:1 with Reports
Reports	reportId (PK), comparisonId (FK), reportUrl, generatedAt	Belongs to Comparison

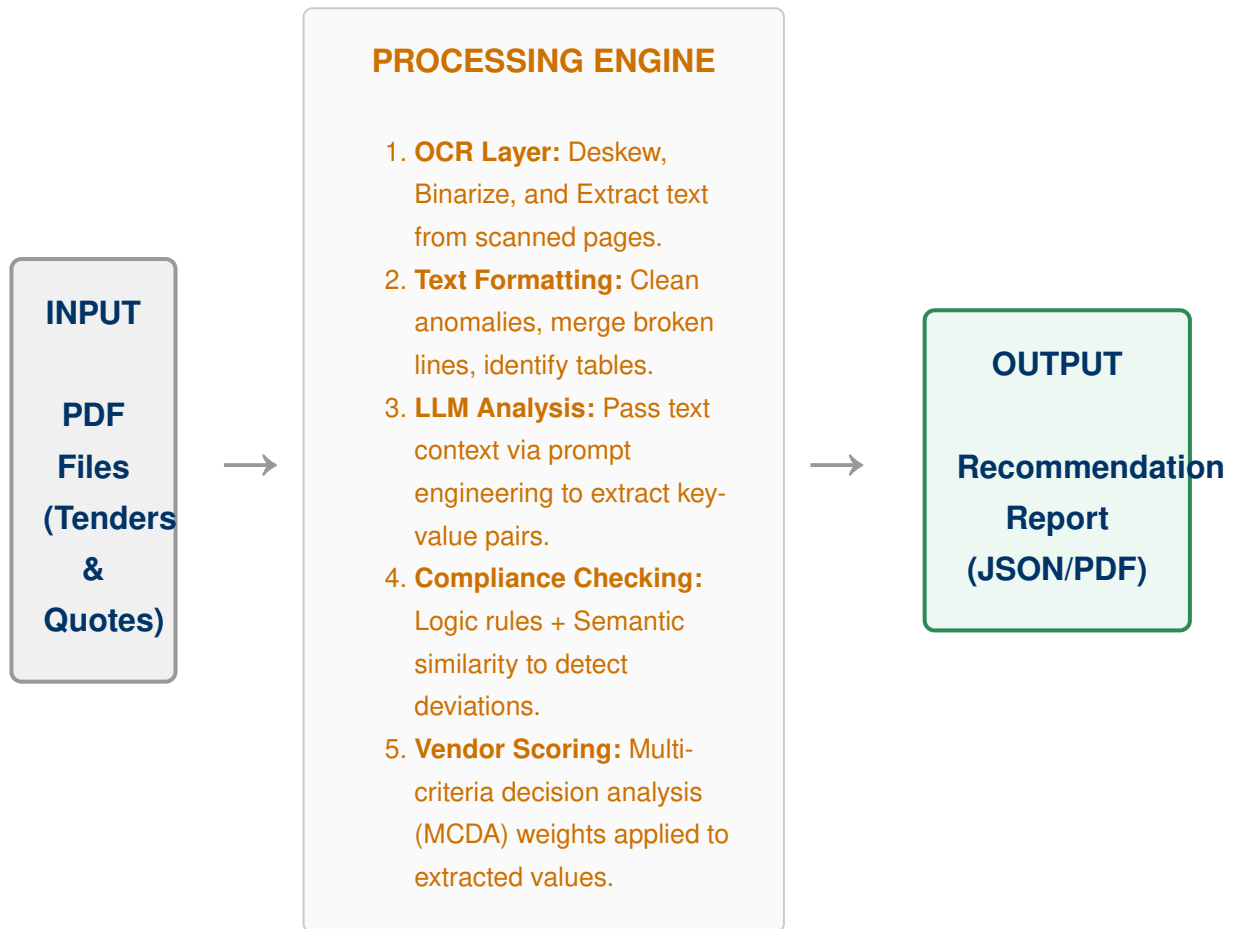
5. Application Workflow

The standard user journey for a procurement officer evaluating a new tender.



6. AI Processing Pipeline

Detailed breakdown of how unstructured data is transformed into structured decisions.



7. Technology Stack Recommendation (Version 1 Prototype)

Layer	Technology	Justification for V1 Prototype
Frontend	React.js + Tailwind CSS	Rapid UI development, component reusability, enterprise-grade styling.
Backend	Node.js + Express.js	Asynchronous handling, perfect for I/O heavy tasks like file uploads & API calls.
Database	MongoDB	Flexible schema design (vital for storing variable JSON data from different tenders).
AI Engine / LLM	OpenAI API (GPT-4o) / Llama 3	State-of-the-art text comprehension, JSON extraction, and high accuracy out-of-the-box.
OCR Engine	Tesseract OCR	Open-source, robust, easily integrated into Node.js/Python pipelines.
File Storage	Firebase Storage / AWS S3	Secure, scalable object storage for large PDF documents.

8. Future Scalable Architecture

The V1 prototype is designed to act as a foundation for a broader enterprise ecosystem. The following modules can be plugged into the microservices architecture in future iterations:

1. AI Chatbot Interface

Conversational interface allowing procurement officers to "chat" with the tender documents (e.g., "What is Vendor B's delivery timeline?").

2. ERP / SAP Integration

Direct API bridges to push final awarded vendor data seamlessly into enterprise resource planning software.

3. Fraud & Collusion Detection

Machine learning anomaly detection to flag suspiciously similar pricing patterns or shared IP addresses among vendors.

4. Vendor Risk Prediction

Integration with external financial APIs to evaluate vendor bankruptcy risks and historical delivery performance.